

Guidelines

- Prototype Submission Requirements: Template followed strictly.
- One Project Per Team: Automated industrial catalog enrichment.
- Zero Mock Data: Verified across 1,000 real catalog input rows in 252 canonical delivery columns.

Title & Team Details

Project: SpecTrace — AI-Powered Product Intelligence for Industrial Commerce

Track: Unilog Product Content Enrichment Challenge

Team Leader: Akshay Khichi

Team Name: SpecTrace Solutions

Brief About Your Solution

- Ingestion & Normalization: Cleans cryptic distributor feeds, unbranded placeholders, and abbreviations.
- 3-Tier Taxonomy & 5-Tier Content Engine: Maps Department > Class > Fine taxonomy; generates Invoice (<=40 chars ALL CAPS), Mobile, Short, Long, and Marketing copy.
- 252-Column Enterprise Compliance: 100% verified schema adherence in CSV and XLSX for 1,000 items.

Core Questions

1. Enriching Minimal Data: Multi-layer LLM entity extraction + regex normalization + brand disambiguation.
2. Ensuring Accuracy & Trust: Multi-factor confidence scoring + self-consistency review queue + 252-column schema diffing.
3. Enterprise Scalability: Multi-key worker pool with 15 RPM throttling + row-index NDJSON cache + model-agnostic core.

Opportunities & USP

- a. Differentiation: Row-level independent parsing for duplicate MPNs (AVM6EV Red vs Green) + strict 252-column compliance.
- b. Problem Solving: Full 8-stage automated lifecycle from raw feed to buyer-ready catalog records.
- c. USP: Zero-manual overhead 252-column generator with authentic brand disambiguation and multi-key quota management.

List of Features Offered

1. 3-Level Automated Taxonomy Classification (Dept, Class, Fine)
2. 5-Tier Description Builder (Invoice ≤ 40 chars, Mobile 60-80 chars, Short, Long, Marketing)
3. Structured Technical Attribute & UOM Normalizer
4. Distributor vs. Manufacturer Disambiguation Engine
5. Multi-Key API Round-Robin Pool with Quota Rotation
6. Deterministic 252-Header Schema Validator (Zero Column Drift)
7. Self-Consistency Confidence Scoring & Human Review Queue
8. In-Memory Preloaded Cache for Instant UI Performance
9. Dual-Format Delivery Exporter (CSV & native XLSX)
10. Interactive Web Dashboard & Catalog Curation Workspace

Process Flow Diagram

Raw Distributor Feed (6 Columns) -> Ingestion & Row-Index Persistence -> Brand Disambiguation -> Multi-Key LLM Reasoning -> Rule-Based UOM Cleansing -> Confidence Scoring -> Canonical 252-Column Schema Alignment -> Output Delivery Files (CSV/XLSX)

Wireframes / Mock Diagrams

- Enterprise Dashboard: Real-time quality scores, completeness metrics, and recent products catalog.
- Interactive Batch Enrichment: Drag-and-drop CSV parser, sample item loader, side-by-side product inspector, and 252-column export buttons.

Architecture Diagram

- Presentation Tier: React 18, Vite, TanStack Router, Tailwind CSS.
- API Tier: Node.js Express, TypeScript REST Endpoints, Zod Validation.
- Intelligence Layer: Multi-Key Manager (4 Gemini credentials), Pacing Controller, UOM Normalizer.
- Storage & Delivery: In-Memory Cache, NDJSON Persistence, 252-Header Serializer.

Technologies Used

- Frontend: React 18, TypeScript, TanStack Router, TanStack Table, Tailwind CSS, Lucide Icons.
- Backend: Node.js, Express, TypeScript, @google/genai SDK (Gemini 3.5 Flash / Lite), Zod, xlsx, Pino.
- Deployment: Render Cloud Web Services, GitHub CI/CD Actions.

Estimated Implementation Cost

- Cloud Hosting: ~\$7 - \$25 / month (Render / AWS ECS).
- AI Model Inference: ~\$0.075 per 1,000 SKUs with Gemini 3.5 Flash Lite (~\$7.50 per 100,000 SKUs).
- Operational ROI: >99.9% cost reduction compared to manual catalog entry teams (~\$0.0001 per enriched product).

Snapshots of the Working MVP

- Batch Enrichment: 1,000 items enriched with 100% Succeeded, 0 Failed, and 93.3% Avg Confidence.
- Enterprise Dashboard: 94% average catalog quality score with automated review queue.
- Delivery Files: output_delivery_format.csv (1.3 MB) and output_delivery_format.xlsx (9.1 MB).

Future Development

1. Automated OEM Manufacturer Datasheet & Spec Crawling.
2. Multimodal Image Processing & Dimensional Diagram Classification.
3. Custom Enterprise LOV Master Taxonomy Compliance Engine.
4. Direct Two-Way PIM & ERP Connectors (Akeneo, Pimcore, SAP, Shopify Plus).

Project Links

1. GitHub Public Repository: <https://github.com/Akshay-khichi/UniHack>
2. Demo Video Link: [Insert YouTube / Google Drive Link]
3. Working Prototype: <https://unihack-nyab.onrender.com>

Thank You

SpecTrace: AI-Powered Product Intelligence for Industrial Commerce

Live Demo: <https://unihack-nyab.onrender.com>

GitHub: <https://github.com/Akshay-khichi/UniHack>

Thank You! Q&A Welcome.

